MIL-C-46477B(MU)
12 June 1967
SUPERSEDING
MIL-C-46477A(Ord)
4 April 1962

MILITARY SPECIFICATION
CARTRIDGE, 7.62MM, NATO, TEST, HIGH PRESSURE, M60

1. SCOPE

1.1 This specification covers Cartridge, 7.62MM, NATO, Test, High Pressure, M60 for use in proof testing 7.62mm weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal
Federal Test Method Standard No. 151 - Metals; Test Methods

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-636 - Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1170 - Visual Standards and Comparison Methods for Evaluating Grain Configuration in 7.62mm Cartridge Cases

DRAWINGS

Munitions Command
C7553703 - Cartridge, 7.62mm, NATO, Test, High Pressure, M60
C7643674 - Classification of Cartridge Case Defects
D8595430 - Packing and Marking Cartridges, 7.62mm, NATO, High Pressure Test, M60; Cartons; Box, Ammunition, M19A1; Box, Wirebound

FSC 1305

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2.1 DRAWINGS (Cont'd)

D8595398	- Packing and marking Cartridges, 7.62mm, NATO, High Pressure Test, M60; Cartons; Box, Ammunition, M2A1; Box, Wirebound
IEL-7553703	- Inspection Equipment List for Cartridge, 7.62mm, NATO, Test, High Pressure, M60

PUBLICATION

AMCR 715-505	- Ammunition Ballistic Acceptance Test Methods, Volume 3, Test Procedures for 7.62mm Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on Drawing C7553703 referenced specifications and the following:

3.2 Bullet extraction.- The force required to extract the bullet from the cartridge case shall be not less than 60 pounds.

3.3 Residual stress.- The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Chamber pressure.- The average chamber pressure of the sample cartridges, conditioned at 68° to 72°F, shall be 67,500 pounds per square inch (psi) plus or minus 2,500 psi. The standard deviation of the chamber pressure of the sample cartridges shall be not greater than 3,000 psi.

3.5 Function and casualty.- The cartridge shall function without casualty at ambient temperature.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.6.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears.

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3.7 Grain configuration.- The grain configuration of the sidewall of the finished cartridge case shall fall within the range defined by the grain configuration standards illustrated in MIL-STD-1170, Figures 1 through 6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 300 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

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4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the manufacturer's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the Procuring Agency.

4.3.2 Examination.- Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects shall be as specified in Table I.

Table I

No.	Defect and Method of Inspection	Major	Minor
	Visual 1/		
	Cartridge		
1	Discolored, dirty, oily, smeared		X
2	Corroded; or stained if etched		X
3	Mixed ammunition types	X	
	Case		
4	Round head		X
5	Dent		X
6	Split case 2/	X	
7	Perforated case	X	
8	Draw scratch		X
10	Beveled underside of head		X
11	Case mouth not crimped in cannelure		X
12	Scaly metal		X
13	No chamfer on head (rim)		X
14	Fold		X
15	Wrinkle		X
16	Buckle		X
17	Bulge		X
18	Illegible or missing head stamp		X
19	Defective head		X
20	Defective mouth		X
21	Coating missing	X	
	Bullet		
24	Split bullet jacket		X

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Table I (Cont'd)

NO.	Defect and Method of Inspection	Major	Minor
25	Loose bullet	X	
26	Missing cannelure		X
27	Scaly metal (bullet)		X
29	Exposed steel (clad jacket)		X
31	Defective cannelure		X
	Primer		
32	No primer	X	
33	Cocked primer	X	
34	Inverted primer	X	
35	Loose primer	X	
36	Nicked or dented primer		X
37	No waterproofing (primer pocket joint)		X
38	Defective crimp		X
	Gaging		
39	Total length	X	
40	Cartridge profile failure (requiring more than 20 pounds dead weight to insert in profile and alignment gage)	X	
41	Diameter of extractor groove		X
42	Diameter of head		X
43	Thickness of head		X
44	Length to shoulder datum,	X	
45	Depth of primer	X	
	Weighing		
46	Weight, min.	X	

1/ Refer to MIL-STD-636 for illustration of defects 1 thru 38, with the exception of Defect No. 21. In the event of conflict between Table I of this specification and MIL-STD-636 concerning defect classification, the classification specified in Table I shall apply.

2/ A split in any location and with or without evidence of propellant loss shall be classed as a major defect.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

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4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

Test	No. of Cartridges	Requirement Paragraph
Bullet Extraction 1/	25	3.2
Residual Stress (Mercurous Nitrate) 1/	50	3.3
Chamber Pressure 2/	25	3.4
Function and Casualty 3/	--	3.5
Grain Configuration 1/	10	3.7
Hardness 2/	10	Drawing

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the initial test, a second sample, consisting of double the number of cartridges in the first sample, shall be tested. If any defective cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the initial test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

3/ The lot shall be rejected when function and casualty defects in the firing of the chamber pressure test exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the initial test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantity of cartridges specified for the chamber pressure test shall be fired. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.- Firing defects and acceptance number in a sample shall be as specified in Table III.

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TABLE III

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative (First and second Sample)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Blown primer 2/	0	1
4. Case casualties		
a. Split (longitudinal)	1	2
b. Complete rupture (circumferential)	0	1
c. Partial rupture (circumferential)	1	2

1/ No second sample permitted. Lot shall be rejected.

2/ Primer not in pocket after firing and both head of cartridge case and pocket enlarged and distorted (determined visually).

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridge shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of any type other than high pressure test shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed in IEL-7553703.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Residual stress (Mercurous Nitrate).- The test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.3 Chamber pressure.- The test shall be conducted in accordance with AMCR 715-505, Vol. 3.

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4.4.4 Function and casualty.- The test shall be conducted simultaneously with the chamber pressure test. When testing of a second sample is required, the test shall be conducted in accordance with the procedure prescribed for the chamber pressure test in AMCR 715-505 Vol. 3 Section 7, using the long piston and procedure in accordance with 7.5.2.1 for fouling shots. All reference pertaining to obtaining chamber pressure data shall be disregarded.

4.4.5 Grain configuration.- The test samples shall be prepared and evaluated in accordance with MIL-STD-1170.

4.4.6 Hardness testing.- The bullets shall be extracted, the propellant removed and the primers extracted. Each cartridge case of the sample shall be prepared and placed on the appropriate test fixture for testing in accordance with Federal Test Method Standard No. 151, Method 244.1. The hardness values of the sample cases for each prescribed point shall be computed and charted in accordance with the drawing requirements.

4.4.7 Defect penalty.- If, in the chamber pressure test, a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. The chamber pressure test shall not be penalized, but the acceptance or initial production sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Level A.- The cartridge shall be packed in accordance with Drawing D8595430 or D8595398 or as required by contract.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders should specify the following:

6.1.1 Title, number and date of the Technical Data Package List (TDPL), for the item covered by this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

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6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Provision for the identification and submission of grain configuration photomicrographs prepared in accordance with MIL-STD-1170 where such submission is required.

6.1.6 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Document availability.

6.2.1 Copies of MIL-STD-1170, required for evaluating grain configuration, may be obtained on a need to have basis through the Commanding Officer, U.S. Army Frankford Arsenal, Philadelphia, Pennsylvania 19137.

Custodian:

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Preparing activity:

Army - MU

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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